

Section 12.1

In Exercises 5–8, refer to Figure 21.

5. Find the components of $\mathbf{u}$.

6. Find the components of $\mathbf{v}$.

7. Find the components of $\mathbf{w}$.

8. Find the components of $\mathbf{q}$.

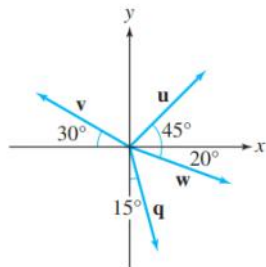


FIGURE 21

Ans 5:-

$$u_x = u \cos \theta = u \cos(45) = u/\sqrt{2}$$

$$u_y = u \sin \theta = u \sin(45) = u/\sqrt{2}$$

Ans 6:-

$$v_x = v \cos(\theta) = v \cos(180 - 30) = -v \frac{\sqrt{3}}{2}$$

$$v_y = v \sin \theta = v \sin(180 - 30) = v/2$$

Ans.-7:-

$$w_x = w \cos(360 - 20) = w \cos 20$$

$$w_y = w \sin(360 - 20) = -w \sin(20).$$

Ans 8:-

$$q_x = q \cos(270 + 15) = q \cos 75$$

$$q_y = q \sin(270 + 15) = -q \sin(75)$$

26. Sketch the vector sum $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3 + \mathbf{v}_4$ in Figure 25(B).

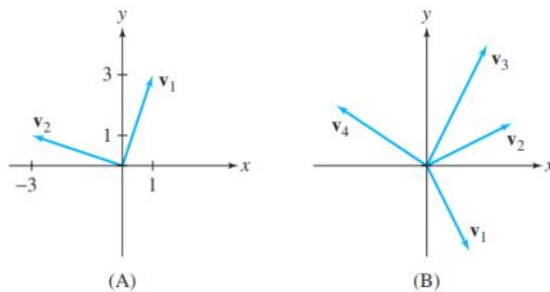


FIGURE 25



64. Determine the magnitude of the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ in Figure 29.

64. Determine the magnitude of the forces F_1 and F_2 in Figure 29, assuming that there is no net force on the object.

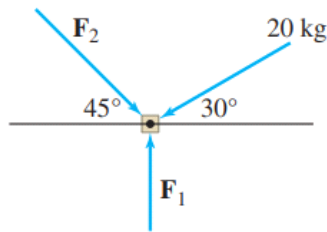


FIGURE 29

$$3^{\text{rd}} \text{ force} = w = mg = 20 \times 9.8 = \boxed{196 \text{ N}}$$

$$x\text{-component of } F_2 = F_2 \cos(45^\circ)$$

$$y\text{-component} = F_2 \sin(45^\circ)$$

$$x\text{-component of } w = 196 \cos(30^\circ)$$

$$y\text{-component of } w = 196 \sin(30^\circ)$$

if net force = 0

hence it is in equilibrium:-

• So, for x -component:-

$$F_2 x = w x$$

$$F_2 \cos(45^\circ) = 196 \cos(30^\circ)$$

$$F_2 = \frac{196 \cos(30^\circ)}{\cos(45^\circ)} = \boxed{240 \text{ N}}$$

• for y -component:-

$$F_1 + F_2 y = w y$$

$$F_1 + F_2 \sin(45^\circ) = 196 \sin(30^\circ)$$

$\therefore$ put the value of F_2 :-

$$F_1 = 196 \sin(30^\circ) - 290 \sin(45^\circ)$$

$$F_1 = 71.70 \text{ N}$$

Section 12.2

In Exercises 24–27, determine whether or not the two vectors are parallel.

26. $\mathbf{u} = \langle 4, 2, -6 \rangle$, $\mathbf{v} = \langle 2, 1, 3 \rangle$

for vectors to be parallel, there must be a scalar multiple:-

$$\mathbf{u} = k\mathbf{v}$$

$$\frac{u_x}{v_x} = \frac{4}{2} = 2 \quad \frac{u_y}{v_y} = \frac{2}{1} = 2 \quad \frac{u_z}{v_z} = \frac{-6}{3} = -2$$

Since the scalar multiple is not constant, hence they are not parallel.

In Exercises 28–31, find the given vector.

31. Unit vector in the direction opposite to $\mathbf{v} = \langle -4, 4, 2 \rangle$

To, find the unit vector opposite to v :-

$$\cdot \quad |v| = \sqrt{(-4)^2 + (4)^2 + (2)^2} = 6$$

• find unit vector:- (opposite)

$$\hat{v} = \frac{-v}{|v|}$$

$$= \left(\frac{4}{6}, \frac{-4}{6}, \frac{-2}{6} \right) = \left\langle \frac{2}{3}, \frac{-2}{3}, \frac{-1}{3} \right\rangle$$

In Exercises 33–40, find a vector parametrization for the line with the given description.

34. Passes through $P = (4, 0, 8)$, direction vector $v = \langle 1, 0, 1 \rangle$

for the vector parametrization:-

$$r(t) = r_0 + tv$$

$$x = 4 + 1t = 4 + t$$

$$y = 0 + 0(t) = 0$$

$$z = 8 + t = 8 + t$$

$$r(t) = \langle 4+t, 0, 8+t \rangle$$

or,

$$x = 4+t, \quad y = 0, \quad z = 8+t$$

40. Passes through $(1, 1, 1)$ parallel to the line through $(2, 0, -1)$ and $(4, 1, 3)$

first we need to find direction vector:-

$$v = (4, 1, 3) - (2, 0, -1) = \langle 2, 1, 4 \rangle$$

Now,

$$r(t) = r_0 + vt$$

$$x = 1 + 2t$$

$$y = 1 + t$$

$$z = 1 + 4t$$

$$r(t) = \langle 1+2t, 1+t, 1+4t \rangle$$

In Exercises 41–44, find parametric equations for the lines with the given description.

42. Perpendicular to the yz -plane, passes through $(0, 0, 2)$

Since it only moves along yz plane :-
hence perpendicular will only move in x -direction.
 $(1, 0, 0)$

$$\begin{array}{lll} x = x_0 + at & y = y_0 + bt & z = z_0 + ct \\ x = t & y = 0 & z = 2 \end{array}$$

$$r(t) = \langle t, 0, 2 \rangle \quad t \in \mathbb{R}$$

48. Show that $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ define the same line, where

$$\mathbf{r}_1(t) = t \langle 2, 1, 3 \rangle, \quad \mathbf{r}_2(t) = \langle -6, -3, -9 \rangle + t \langle 8, 4, 12 \rangle$$

• first we check their direction:-

$$\begin{aligned} \mathbf{r}_1(t) &= \langle 2, 1, 3 \rangle \\ \mathbf{r}_2(t) &= \langle 8, 4, 12 \rangle \end{aligned}$$

$$k = \frac{8}{2} = 4$$

$$k = \frac{4}{1} = 4$$

$$k = \frac{12}{3} = 4$$

as the scalar constant is same (ie. 4), so they are in same direction.

Now, we will check the points:-

at $t=0$ the point lies on r_2 is $(-6, -3, -9)$

for at $t=-3$ the point lies on $r_1 = (2x-3, 1x-3, 3x-3)$
 $= (-6, -3, -9)$

Here the point exists on both lines and both equations represent the same line.

52. Determine whether the lines $\mathbf{r}_1(t) = \langle 2, 1, 1 \rangle + t \langle -4, 0, 1 \rangle$ and $\mathbf{r}_2(s) = \langle -4, 1, 5 \rangle + s \langle 2, 1, -2 \rangle$ intersect, and if so, find the point of intersection.

To check whether the lines intersect or not:-
if they intersect that means:-
 $r_1(t) = r_2(s)$

• Parametrization:-

$$r_1 x = 2 + (-4t) = 2 - 4t$$

$$r_1 y = 1 + 0t = 1$$

$$r_1 z = 1 + t$$

$$r_2 x = -4 + 2s$$

$$r_2 y = 1 + s$$

$$r_2 z = 5 - 2s$$

- equating for x :-

$$2 - 4t = -4 + 2s$$

$$3 = 2t + s$$

$$s = 3 - 2t \text{ --- eq(ii)}$$

- equation for y :-

$$1 = 1 + s$$

$$\boxed{s = 0}$$

put in eq(i) :-

$$0 = 3 - 2t$$

$$\boxed{t = 3/2}$$

- equate both the values for z :-

$$1 + t = 5 - 2s$$

$$1 + 3/2 = 5 - 0$$

$$5/2 \neq 5 \quad \text{Hence they don't intersect.}$$

Section 12.3:-

29. Find all values of b for which the vectors are orthogonal.

(a) $\langle b, 3, 2 \rangle, \langle 1, b, 1 \rangle$

(b) $\langle 4, -2, 7 \rangle, \langle b^2, b, 0 \rangle$

- Two vectors are orthogonal if their dot product is zero

a)

$$b(1) + 3b + 2 = 0$$

$$4b + 2 = 0$$

$$b = -2/4 = \boxed{-1/2}$$

b)

$$4b^2 - 2b + 0 = 0$$

$$\boxed{b = 1/2, 0}$$

32. Find a vector that is orthogonal to $\mathbf{v} = \langle 1, 2, 1 \rangle$ but not to $\mathbf{w} = \langle 1, 0, -1 \rangle$.

- a vector to be orthogonal to $\mathbf{v}$:-

$$\mathbf{u} \cdot \mathbf{v} = 0$$

$$\langle x, y, z \rangle \cdot \langle 1, 2, 1 \rangle = x + 2y + z = 0$$

$$\text{for } x=0 \quad y=1 \quad z=-2$$

$$0 + 2 - 2 = 0$$

$$\boxed{0 = 0}$$

- but the vector should not be orthogonal to $\mathbf{w}$:-

$$\mathbf{u} \cdot \mathbf{w} \neq 0$$

$$\langle 0, 1, -2 \rangle \cdot \langle 1, 0, -1 \rangle = 0 + 0 + 2 = \boxed{2 \neq 0}$$

Here the vector is $\langle 0, 1, -2 \rangle$

48. Find $\|2\mathbf{e} - 3\mathbf{f}\|$, assuming that $\mathbf{e}$ and $\mathbf{f}$ are unit vectors such that $\|\mathbf{e} + \mathbf{f}\| = \sqrt{3/2}$.

$$\bullet \quad (a-b)^2 = a^2 + b^2 - 2ab$$

$$\|2\mathbf{e} - 3\mathbf{f}\|^2 = \|2\mathbf{e}\|^2 + \|-3\mathbf{f}\|^2 - 2((2\mathbf{e})(-3\mathbf{f}))$$

$$\|2\mathbf{e} - 3\mathbf{f}\|^2 = 4|\mathbf{e}|^2 + 9|\mathbf{f}|^2 - 12(\mathbf{e} \cdot \mathbf{f})$$

$$\text{since } \mathbf{e}, \mathbf{f} \text{ are unit vectors } e^2, f^2 = 1$$

$$= 4 + 9 - 12(\mathbf{e} \cdot \mathbf{f})$$

• To find $\mathbf{e} \cdot \mathbf{f}$:-

$$- \quad (a+b)^2 = a^2 + b^2 + 2ab$$

$$\|\mathbf{e} + \mathbf{f}\|^2 = |\mathbf{e}|^2 + |\mathbf{f}|^2 + 2(\mathbf{e} \cdot \mathbf{f})$$

$$(\sqrt{3/2})^2 = 2 + 2(\mathbf{e} \cdot \mathbf{f})$$

$$\frac{3}{2} = 2 + 2(\mathbf{e} \cdot \mathbf{f})$$

$$\boxed{\mathbf{e} \cdot \mathbf{f} = -5/8}$$

• Now,

$$|2e-3f|^2 = 13 + 12\left(\frac{-5}{8}\right)$$

$$|2e-3f| = \sqrt{\frac{11}{2}} \quad \text{Ans.}$$

64. Find $\|\mathbf{u}_{\perp v}\|$ in Figure 15.

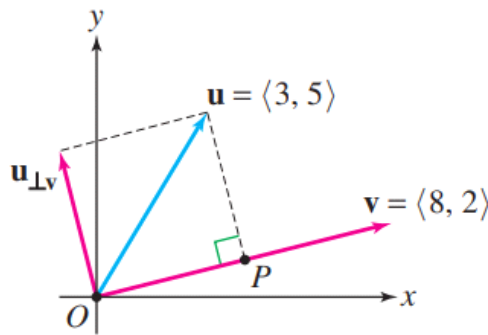


FIGURE 15

$$\mathbf{u}_{\perp v} = \mathbf{u} - \mathbf{u}_{\parallel v}$$

$$u_{\parallel v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|} \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\langle 3, 5 \rangle \cdot \langle 8, 2 \rangle}{8^2 + 2^2} = \frac{24 + 10}{68} = \frac{34}{68} \cdot \mathbf{v}$$

$$\mathbf{u}_{\perp v} = \mathbf{u} - \mathbf{u}_{\parallel v} = \langle 3, 5 \rangle - \frac{34}{68} \langle 8, 2 \rangle = \langle 3, 5 \rangle - \langle 4, 1 \rangle$$

$$= \langle -1, 4 \rangle$$

for magnitude:- $\sqrt{(-1)^2 + 4^2} = \sqrt{17} \quad \text{Ans}$

In Exercises 73–76, refer to Figure 16.

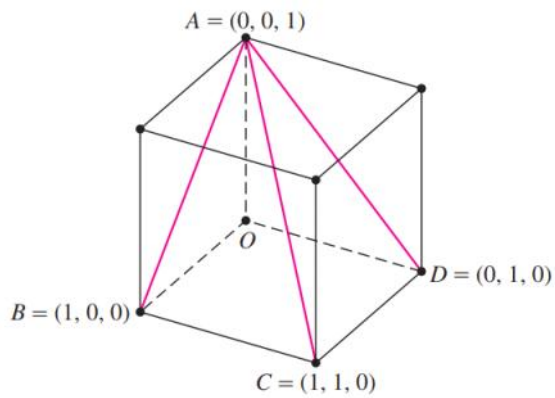


FIGURE 16 Unit cube in $\mathbf{R}^3$.

73. Find the angle between $\overline{AB}$ and $\overline{AC}$.

$$\overline{AB} = B - A = (1, 0, 0) - (0, 0, 1)$$

$$\overline{AB} = \langle 1, 0, -1 \rangle$$

$$\overline{AC} = C - A = (1, 1, 0) - (0, 0, 1)$$

$$\overline{AC} = \langle 1, 1, -1 \rangle$$

angle between $\overline{AB}$ & $\overline{AC}$:-

$$\overline{AB} \cdot \overline{AC} = |\overline{AB}| |\overline{AC}| \cos \theta$$

$$\langle 1, 0, -1 \rangle \cdot \langle 1, 1, -1 \rangle = (\sqrt{1^2 + 1^2}) (\sqrt{1^2 + 1^2 + 1^2}) \cos \theta$$

$$\frac{(1+0+1)}{\sqrt{2} \cdot \sqrt{3}} = \cos \theta$$

$$\frac{\sqrt{2}}{\sqrt{3}} = \cos Q \quad Q = \cos^{-1}\left(\sqrt{2/3}\right)$$

$$\boxed{Q = 35.26^\circ}$$

74. Find the angle between $\overline{AB}$ and $\overline{AD}$.

$$\overline{AB} = \langle 1, 0, -1 \rangle$$

$$\overline{AD} = D - A = (0, 1, 0) - (0, 0, 1) = \langle 0, 1, -1 \rangle$$

$$\overline{AB} \cdot \overline{AD} = |\overline{AB}| |\overline{AD}| \cos Q$$

$$\langle 1, 0, -1 \rangle \cdot \langle 0, 1, -1 \rangle = (\sqrt{1^2 + 1^2})(\sqrt{1^2 + 1^2}) \cos Q$$

$$(1 + 0 + 1) = (\sqrt{2})(\sqrt{2}) \cos Q$$

$$2 = 2 \cos Q$$

$$Q = \cos^{-1}(1) = 0^\circ$$

75. Calculate the projection of $\overrightarrow{AC}$ along $\overrightarrow{AD}$.

$$\text{Projection of AC on AD} = \frac{\overline{AC} \cdot \overline{AD}}{|\overline{AD}|} \cdot \overline{AD}$$

$$= \langle 1, 1, -1 \rangle \cdot \langle 0, 1, -1 \rangle$$

$$= \frac{\langle 1, 1, -1 \rangle \cdot \langle 0, 1, -1 \rangle}{\sqrt{1^2 + 1^2}}$$

$$= \frac{(0 + 1 + 1)}{\sqrt{2}} \cdot \langle 0, 1, -1 \rangle$$

$$= \sqrt{2} \langle 0, 1, -1 \rangle = \boxed{\langle 0, \sqrt{2}, -\sqrt{2} \rangle \text{ Ans}}$$

76. Calculate the projection of $\overrightarrow{AD}$ along $\overrightarrow{AB}$.

$$\overrightarrow{AD} = \langle 0, 1, -1 \rangle$$

$$\overrightarrow{AB} = \langle 1, 0, -1 \rangle$$

$$\text{projection of } \overrightarrow{AD} \text{ on } \overrightarrow{AB} = \frac{\overrightarrow{AD} \cdot \overrightarrow{AB}}{|\overrightarrow{AB}|} \cdot \overrightarrow{AB}$$

$$= \frac{\langle 0, 1, -1 \rangle \cdot \langle 1, 0, -1 \rangle}{\sqrt{1^2 + 1^2}} \cdot \langle 1, 0, -1 \rangle$$

$$= \frac{1}{\sqrt{2}} \cdot \langle 1, 0, -1 \rangle$$

$$= \boxed{\left\langle \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}} \right\rangle \text{ Ans.}}$$

85. Prove that $\|v + w\|^2 - \|v - w\|^2 = 4v \cdot w$.

86. Use Exercise 85 to show that v and w are orthogonal if and only if $\|v - w\| = \|v + w\|$.

as we know if $v \perp w$ are orthogonal then,
 $v \cdot w = 0$

so,

lets say Q85 was proved!
then put the value of $v \cdot w$:-

$$|v+w|^2 - |v-w|^2 = 4v \cdot w$$

$$|v+w|^2 - |v-w|^2 = 4(0)$$

$$|v+w|^2 = |v-w|^2$$

Taking root on b.s:-

$$\boxed{|v+w| = |v-w|}, \text{ hence proved!}$$

Section 12.4

26. Which of the following form a right-handed system?

- (a) $\{v, w, u\}$ (b) $\{w, v, u\}$ (c) $\{v, u, w\}$
(d) $\{u, v, w\}$ (e) $\{w, v, -u\}$ (f) $\{v, -u, w\}$

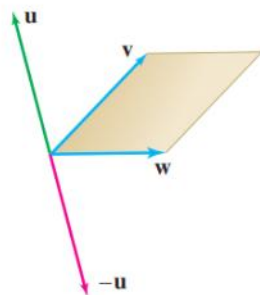


FIGURE 17

b, c are right handed systems.

38. Calculate the volume of the parallelepiped spanned by

$$\mathbf{u} = \langle 2, 2, 1 \rangle, \quad \mathbf{v} = \langle 1, 0, 3 \rangle, \quad \mathbf{w} = \langle 0, -4, 0 \rangle$$

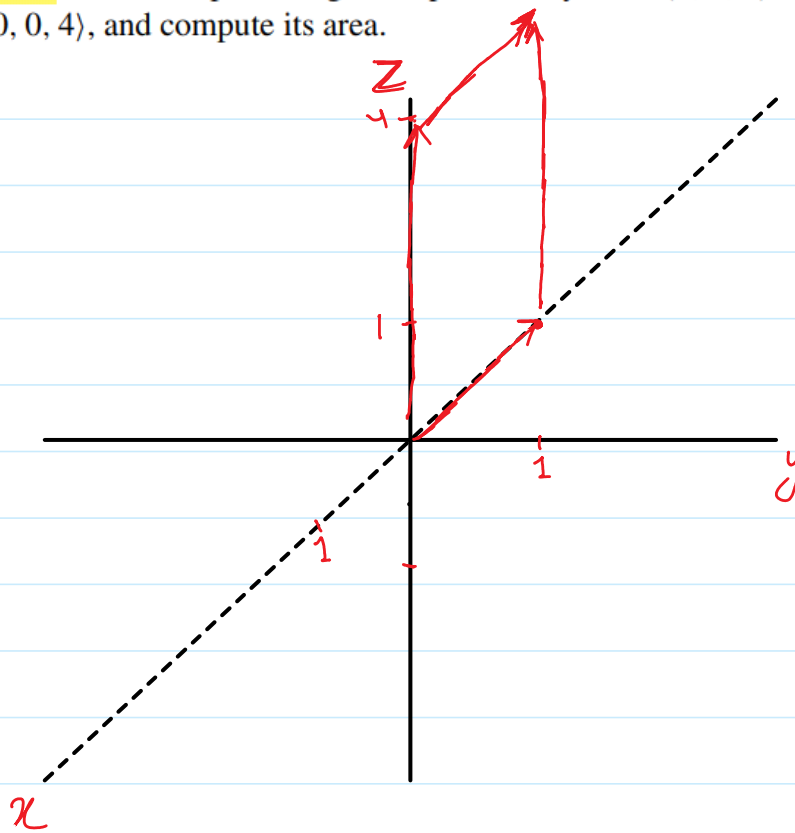
$$\text{Volume of parallelepiped} = |\vec{u} \ \vec{v} \ \vec{w}|$$

$$= \begin{vmatrix} 2 & 2 & 1 \\ 1 & 0 & 3 \\ 0 & -4 & 0 \end{vmatrix}$$

$$= 2 \begin{vmatrix} 0 & 3 \\ -4 & 0 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ -4 & 0 \end{vmatrix}$$

$$= 2(0 + 12) - 1(0 + 4) \\ = 24 - 4 = 20 \text{ unit}^3$$

40. Sketch the parallelogram spanned by $\mathbf{u} = \langle 1, 1, 1 \rangle$ and $\mathbf{v} = \langle 0, 0, 4 \rangle$, and compute its area.



area of parallelogram = $|\mathbf{A} \times \mathbf{B}| = |\mathbf{u} \times \mathbf{v}|$

$$|\mathbf{u} \times \mathbf{v}| = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 0 & 0 & 4 \end{vmatrix}$$

$$= i \begin{vmatrix} 1 & 1 \\ 0 & 4 \end{vmatrix} - j \begin{vmatrix} 1 & 1 \\ 0 & 4 \end{vmatrix} + k \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}$$

$$= i(4-0) - j(4-0) + k(0)$$

$$\mathbf{u} \times \mathbf{v} = 4i - 4j$$

$$|\mathbf{u} \times \mathbf{v}| = \sqrt{4^2 + 4^2} = \boxed{\sqrt{32} \text{ unit}^2 \text{ Area}}$$

44. Use the cross product to find the area of the triangle with vertices $P = (1, 1, 5)$, $Q = (3, 4, 3)$, and $R = (1, 5, 7)$ (Figure 20).

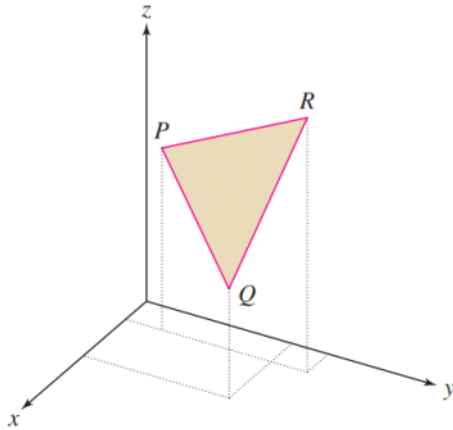


FIGURE 20

$$\text{Area of triangle} = \frac{1}{2} |A \times B|$$

$$A = \overrightarrow{PQ} = Q - P = (3, 4, 3) - (1, 1, 5)$$

$$\vec{A} = \langle 2, 3, -2 \rangle$$

$$B = \overrightarrow{PR} = R - P = (1, 5, 7) - (1, 1, 5)$$

$$\vec{B} = \langle 0, 4, 2 \rangle$$

$$\text{Area} = \frac{1}{2} |A \times B| = \frac{1}{2} \begin{vmatrix} i & j & k \\ 2 & 3 & -2 \\ 0 & 4 & 2 \end{vmatrix}$$

$$= \frac{1}{2} \left[i \begin{vmatrix} 3 & -2 \\ 4 & 2 \end{vmatrix} - j \begin{vmatrix} 2 & -2 \\ 0 & 2 \end{vmatrix} + k \begin{vmatrix} 2 & 3 \\ 0 & 4 \end{vmatrix} \right]$$

$$= \frac{1}{2} \begin{vmatrix} 1 & 4 & 2 \\ 1 & 0 & 2 \\ 1 & 0 & 4 \end{vmatrix}$$

$$= \frac{1}{2} [i(6+8) - j(4-0) + k(8-0)]$$

$$= \frac{1}{2} [14i - 4j + 8k]$$

$$= \frac{1}{2} \sqrt{14^2 + 4^2 + 8^2}$$

$$= \frac{1}{2} (16.6) = \boxed{8.30 \text{ unit}^2}$$

57.  Show that three points P, Q, R are collinear (lie on a line) if and only if $\vec{PQ} \times \vec{PR} = \mathbf{0}$.

$$\vec{PQ} = Q - P$$

$$\vec{PR} = R - P$$

- The points are collinear only if they are at the same line that means the angle between them [ie $\theta = 0$].

So, we know that:-

$$|\vec{PQ} \times \vec{PR}| = |\vec{PQ}| |\vec{PR}| \sin \theta$$

put the value of $\theta = 0$:-

$$PQ \times PR = |PQ| |PR| \sin(\theta) = 0$$

Here proved.

58. Use the result of Exercise 57 to determine whether the points P , Q , and R are collinear, and if not, find a vector normal to the plane containing them.

(a) $P = (2, 1, 0)$, $Q = (1, 5, 2)$, $R = (-1, 13, 6)$

(b) $P = (2, 1, 0)$, $Q = (-3, 21, 10)$, $R = (5, -2, 9)$

(c) $P = (1, 1, 0)$, $Q = (1, -2, -1)$, $R = (3, 2, -4)$

a) $|PQR| = 0$

$$= \begin{vmatrix} 2 & 1 & 0 \\ 1 & 5 & 2 \\ -1 & 13 & 6 \end{vmatrix} = 2 \begin{vmatrix} 5 & 2 \\ 13 & 6 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ -1 & 6 \end{vmatrix}$$

$$= 2(30 - 26) - 1(6 + 2)$$

$$= 8 - 8 = 0$$

Here the points are collinear.

b) $\begin{vmatrix} 2 & 1 & 0 \\ -3 & 21 & 10 \\ 5 & -2 & 9 \end{vmatrix} = 2 \begin{vmatrix} 21 & 10 \\ -2 & 9 \end{vmatrix} - 1 \begin{vmatrix} -3 & 10 \\ 5 & 9 \end{vmatrix}$

$$= 2(189 + 20) - 1(-27 - 50)$$

$$= 418 + 77 = 495$$

points are not collinear.

- finding normal-

$$\overrightarrow{PQ} = Q - P = (-3, 21, 10) - (2, 1, 10) \\ = \langle -5, 20, 0 \rangle$$

$$\overrightarrow{PR} = R - P = (5, -2, 9) - (2, 1, 10) \\ = \langle 3, -3, -1 \rangle$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} i & j & k \\ -5 & 20 & 0 \\ 3 & -3 & -1 \end{vmatrix} = i \begin{vmatrix} 20 & 0 \\ -3 & -1 \end{vmatrix} - j \begin{vmatrix} -5 & 0 \\ 3 & -1 \end{vmatrix} + k \begin{vmatrix} -5 & 20 \\ 3 & -3 \end{vmatrix} \\ = i(-20 - 0) - j(5 - 0) + k(15 - 60) \\ = -20i - 5j - 45k$$

$$\text{vector normal to the plane} = \langle -20, -5, -45 \rangle$$

$$c) \begin{vmatrix} 1 & 1 & 0 \\ 1 & -2 & -1 \\ 3 & 2 & -4 \end{vmatrix} = 1 \begin{vmatrix} -2 & -1 \\ 2 & -4 \end{vmatrix} - 1 \begin{vmatrix} 1 & -1 \\ 3 & -4 \end{vmatrix} \\ = 1(8 + 2) - 1(-4 + 3) \\ = 10 + 1 = 11$$

$$\overrightarrow{PQ} = Q - P = (1, -2, 1) - (1, 1, 0) = (0, -3, 1)$$

$$\overrightarrow{PR} = R - P = (3, 2, -4) - (1, 1, 0) = (2, 1, -4)$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} i & j & k \\ 0 & -3 & 1 \\ 2 & 1 & -4 \end{vmatrix} = i \begin{vmatrix} -3 & 1 \\ 1 & -4 \end{vmatrix} - j \begin{vmatrix} 0 & 1 \\ 2 & -4 \end{vmatrix} + k \begin{vmatrix} 0 & -3 \\ 2 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 0 & -1 & 1 \\ 2 & 1 & 4 \end{vmatrix} = i(-12+1) - j(0+2) + k(0+6) \\ = -11i - 2j + 6k$$

Vector normal to the plane = $\langle -11, -2, 6 \rangle$

Section 12.5

In Exercises 21–28, find the equation of the plane with the given description.

24. Passes through $P = (3, 5, -9)$ and is parallel to the xz -plane

Since it is parallel to xz -plane, the equation won't involve x and z axis:-

$$ax + by + cz = d$$

$$\boxed{y = 5}$$

26. Contains the lines $\mathbf{r}_1(t) = \langle t, 2t, 3t \rangle$ and $\mathbf{r}_2(t) = \langle 3t, t, 8t \rangle$

confusion.

finding direction vectors:-

$$\mathbf{r}_1(t) = \frac{d\mathbf{r}}{dt} \langle t, 2t, 3t \rangle = \langle 1, 2, 3 \rangle = \mathbf{a}$$

$$r_2(t) = \frac{dr_1}{dt} (3t, t, 8t) = (3, 1, 8) = b$$

find normal to the plane.

$$\begin{aligned} a \times b &= \begin{vmatrix} i & j & k \\ 1 & 2 & 3 \\ 3 & 1 & 8 \end{vmatrix} = i \begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} - j \begin{vmatrix} 1 & 3 \\ 3 & 8 \end{vmatrix} + k \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \\ &= i(16-3) - j(8-9) + k(1-6) \\ &= i(13) - j(-1) + k(-5) \end{aligned}$$

$$\text{normal} = \langle 13, 1, -5 \rangle$$

- find the equation of the plane:-
let's take reference point $P_0 = (0, 0, 0)$

$$n_1(x-x_0) + n_2(y-y_0) + n_3(z-z_0) = 0$$

$$13(x-0) + 1(y-0) - 5(z-0) = 0$$

$$\boxed{13x + y - 5z = 0 \quad \text{Ans}}$$

30. Are the planes $2x - 4y - z = 3$ and $-6x + 12y + 3z = 1$ parallel?

$$\begin{aligned} \text{let } a &= 2x - 4y - z = 3 \\ b &= -6x + 12y + 3z = 1 \end{aligned}$$

$$\text{normal of } a = \langle 2, -4, -1 \rangle$$

normal of $b = \langle -6, 12, 3 \rangle$

- If they are parallel, they must be scalar multiple of each other.

solving:-

$$\frac{b_1}{a_1} = \frac{-6}{2} = -3$$

$$\frac{b_2}{a_2} = \frac{12}{-4} = -3$$

$$\frac{b_3}{a_3} = \frac{3}{-1} = -3$$

hence $n_1 = k n_2$ where $\boxed{k = -3}$

planes are parallel!

In Exercises 39–42, find the intersection of the line and the plane.

42. $x - z = 6$, $\mathbf{r}(t) = \langle 1, 0, -1 \rangle + t \langle 4, 9, 2 \rangle$

equation of the line:- $\mathbf{r}(t)$

parametrization

$$\begin{aligned} x &= 1 + 4t && \text{--- } I \\ y &= 9t && \text{--- } II \\ z &= -1 + 2t && \text{--- } III \end{aligned}$$

put in equation of plane-

$$(1 + 4t) - (-1 + 2t) = 6$$

$$1 + 4t + 1 - 2t = 6$$

$$2 + 2t = 6$$

$$2t = 4$$

$$\boxed{t = 2}$$

Put the value of t in eq i, ii, iii for point of intersection.

$$x = 1 + 4 \times 2 = 9$$

$$y = 9 \times 2 = 18$$

$$z = -1 + 4 = 3$$

intersection point = $(9, 18, 3)$

50. Give equations for two distinct planes whose trace in the xy -plane has equation $4x + 3y = 8$.

To form actual plane equations in three dimensions, we can introduce a **z-term** while keeping the x - and y -terms unchanged.

Simplest choice are any non zero numbers:-

$$2z \rightarrow 4x + 3y + 2z = 8$$

Answer!

$$-5z \rightarrow 4x + 3y - 5z = 8$$

53. Find all planes in $\mathbf{R}^3$ whose intersection with the xz -plane is the line with equation $3x + 2z = 5$.

Since the plane must still satisfy the equation $3x+2z=5$ when $y=0$, but can extend in the y -direction, we add a By -term

$$3x + By + 2z = 5 \quad \text{for } B \in \mathbb{R}$$

This represents infinite family of planes that all pass through the given equation of xz -plane.

66. Find parametric equations for the intersection of the planes $2x + y - 3z = 0$ and $x + y = 1$.

$$\begin{aligned} x + y &= 1 \\ y &= 1 - x \end{aligned}$$

Put in eq (i):-

$$2x + (1 - x) - 3z = 0$$

$$2x + 1 - x - 3z = 0$$

$$x + 1 = 3z$$

$$x = 3z - 1$$

Let $z = t$ (free variable)

$$x = 3t - 1$$

$$y = 1 - (3t - 1)$$

$$y = 2 - 3t$$

$$z = t$$

Thus, parametric equations are :-

$$x = 3t - 1, \quad y = 2 - 3t, \quad z = t \quad \text{for } t \in \mathbb{R}$$

68. The plane

$$\frac{x}{2} + \frac{y}{4} + \frac{z}{3} = 1$$

intersects the x -, y -, and z -axes in points P , Q , and R . Find the area of the triangle $\triangle PQR$.

• first find the intercepts:-

$$- \quad \frac{x}{2} = 1 \quad x = 2$$

$$P(2, 0, 0)$$

$$- \quad \frac{y}{4} = 1 \quad y = 4$$

$$Q(0, 4, 0)$$

$$- \quad \frac{z}{3} = 1 \quad z = 3$$

$$R(0, 0, 3)$$

$$• \quad \overrightarrow{PQ} = Q - P = (-2, 4, 0)$$

$$• \quad \overrightarrow{PR} = R - P = (-2, 0, 3)$$

- $\vec{r}_K = \vec{K} - \vec{P} = (-2, 0, 3)$

- $\text{Area} = |\vec{PQ} \times \vec{PR}| = \begin{vmatrix} i & j & k \\ -2 & 4 & 0 \\ -2 & 0 & 3 \end{vmatrix}$

$$= i \begin{vmatrix} 4 & 0 \\ 0 & 3 \end{vmatrix} - j \begin{vmatrix} -2 & 0 \\ -2 & 3 \end{vmatrix} + k \begin{vmatrix} -2 & 4 \\ -2 & 0 \end{vmatrix}$$

$$= i(12) - j(-6 + 0) + k(0 + 8)$$

$$= 12i + 6j + 8k$$

$$\text{Area} = \sqrt{12^2 + 6^2 + 8^2} \quad \boxed{= 15.62 \text{ unit}^2}$$